

System Design Document

furlink

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Introduction

This document outlines the technical architecture for furlink. furlink is designed to replace a manual booking process with an integrated digital ecosystem. The system centralizes pet registration, scheduling, and payment processing into a single platform accessible via the web.

Purpose

The purpose of this document is to ensure the furlink design meets established project requirements for real-time booking and business management. It provides a detailed description of the system architecture, software modules, and security protocols to ensure organizational goals for efficient pet care management are achieved.

System Overview

This section describes the core architectural goals and operational approaches of furlink. Designed to replace manual, fragmented coordination methods that frequently lead to scheduling conflicts and inconsistent record management, furlink standardizes the digital marketplace experience to optimize scheduling efficiency, data visibility, and workflow transparency for both pet owners and service providers.

The system is designed to achieve four primary goals:

1. **Ability to Customize Pet Haircut:** Successfully develop and deploy an AI-powered haircut preview tool, integrated into the platform, by August 2026, ensuring it achieves a minimum of 70% image rendering accuracy, accurate identification of selected haircut suitability for the pet's current hair style and condition.
2. **Reduce Scheduling Conflicts:** Eliminate double booking by ensuring that each service provider does not experience booking incidents during the first three (3) months of the release by integrating an AI Booking Assistant.

3. **Improve Visibility of Pet Service Providers:** Improve the visibility of small to medium-sized pet care service providers by giving them a dedicated platform to showcase their services and directly connect pet owners. Onboard and verify one (1) service provider within the first three (3) months of release.
4. **Provide Business Insights for Service Providers:** Enable pet care service providers to track monthly sales, customer feedback, and related insights by integrating a business analytics dashboard and AI-powered Data Analyst into the platform by August 2026.

The furlink architecture utilizes a centralized, shared cloud database engine to ensure all operational datasets remain fully synchronized and consistent across modules in real time. The web presentation layer is engineered using React.js hosted on Vercel, with Supabase leveraging a cloud-native PostgreSQL instance to serve as the secure backend-as-a-service (BaaS) and database repository. Furthermore, PayMongo is securely integrated via API endpoints as the official external payment clearing gateway, while Supabase Edge Functions orchestrate all serverless background processing, including distributed AI model execution, Gmail SMTP transactional notifications, and automated payment webhook handshakes.

Design Constraints

The engineering and implementation of the furlink platform are bound by several critical technical, operational, and regulatory constraints. These boundaries originate outside the immediate application scope but must be structurally factored into the system architecture to ensure predictable financial runtimes, platform stability, and compliance.

1. Hardware and Technology Constraints

- **Continuous Cloud Dependency:** The platform functions as a purely cloud-native, serverless architecture. It possesses a non-negotiable requirement for an active, low-latency internet connection to execute core operations, including live marketplace listings, asynchronous slot scheduling, data schema synchronization, and secure transaction workflows.

- **Serverless Compute Cold Starts:** Backend functions deployed via cloud layers are bound by cold start constraints. Periods of platform inactivity can introduce a transient processing latency of 1 to 3 seconds during initial invocation, impacting immediate API responsiveness, real-time message routing, and background webhook processing.
- **AI Rate Limiting:** To mitigate the high risk of AI API Cost Overrun and protect the team's budget allocation, strict programmatic rate limits are enforced at the backend middleware layer. While concurrent traffic interacting with the AI Booking Assistant, AI Pet Haircut Generator, or AI-Powered Data Analyst could cause transaction queuing or delayed execution if user caps are met, this throttling design constraint is a deliberate financial guardrail.
- **Generative Image Latency and Dual-Provider Fallback:** The AI Pet Haircut Generator is constrained by heavy GPU compute cycles required to render high-fidelity graphics. To balance latency, computational availability, and resource expenditure, the system employs a tiered configuration strategy utilizing Hugging Face Serverless Inference as the primary image engine. If Hugging Face encounters transient request timeouts or configured image generation fallback, meaning users may experience a variable rendering latency of several seconds depending on the active provider.
- **API Service Availability:** The text-based intelligent modules—specifically the AI Booking Assistant and the AI-powered Data Analyst—rely exclusively on the Google Gemini 2.5 Flash API with no secondary fallback model engineered into the pipeline. This single-model dependency is justified by two architectural realities: first, real-world telemetry confirms that Google's enterprise infrastructure maintains an exceptional 99.9% operational availability, rendering its cumulative 8 hours and 45 minutes annual distributed downtime an acceptable operational risk; second, because conversational booking guidance and analytical report parsing represent non-critical enhancements rather than core utilities, any transient Gemini outage

will not disrupt the platform's primary database booking engine or PayMongo clearing workflows. If an outage occurs, the frontend gracefully degrades to standard, rule-based database pathing without compromising application uptime.

2. Software and System Integration

- **Two-Tiered Booking Finalization and Webhook Coupling:** The finalization of a pet grooming appointment is strictly decoupled into a two-tiered state machine. The system design is constrained such that receiving a successful payment confirmation signal from the external PayMongo webhook engine does not automatically finalize a booking; it merely transitions the database record into a secured, paid request state. Absolute finalization and database state persistence of an appointment are legally and programmatically coupled with a secondary, manual review and acceptance signal triggered by the designated Service Provider. Any third-party webhook drop or provider connectivity lag directly holds the transaction in an intermediate reservation queue.
- **Role-Based Access Control:** The application layout enforces strict data isolation across three distinct user dimensions: Pet Owners, Service Providers, and Application Administrators. This structural constraint requires separate permission profiles and Row-Level Security (RLS) tokens to be securely evaluated across all serverless endpoints to prevent lateral data privilege escalation.
- **AI Context Window Dependency:** The continuous contextual awareness of the AI Booking Assistant and the AI-Powered Data Analyst is constrained by the mathematical token boundaries of the Gemini context window. Extended chat sessions that accumulate history beyond this memory parameter require the system to drop oldest messages or prompt the user to re-input parameters to prevent processing errors or conversational context loss.

3. Standards and Legal Rules

- **Data Privacy Act (R.A. 10173) Compliance:** To comply with the Philippine Data Privacy Act of 2012, all stored personal, pet health, and commercial transaction records must adhere to strict protection, minimization, and right-to-erasure mandates. The AI Booking Assistant is programmatically restricted from accessing any user data outside of what the user has explicitly typed or stored in their active session. Similarly, the AI-Powered Data Analyst is sandboxed to compile metrics strictly from aggregated, anonymized business data belonging to the authenticated service provider, completely blocking cross-tenant data leaks.
- **Payment Architecture Decoupling:** To align with global digital payment security standards, the platform is strictly prohibited from capturing, processing, or caching financial card data or bank credentials within its internal database. The AI Booking Assistant's capabilities are legally restricted to generating encrypted PayMongo checkout strings, completely transferring the transactional data-entry phase onto PayMongo's isolated, compliant external environment.
- **Pet Care Operational Workflows:** The system must strictly enforce standard veterinary and pet care compliance standards. This includes mandatory verification parameters for uploaded vaccination histories, age boundaries for specific treatments, and illness disclosure fields before a grooming appointment request can be forwarded to a provider.
- **Transactional Audit Accountability:** A permanent, immutable audit trail must be recorded automatically for every commercial booking transaction, escrowed payment state, provider acceptance decision, and administrative change request. This logging constraint ensures total platform transparency and operational accountability in line with standard consumer service industry practices.

Roles and Responsibilities

The table below defines furlink System Design roles and responsibilities and serves as the list of points of contact for issues and concerns:

Name	Role	Email
Lalaine Andaya	Project Manager	lcandaya2@student.apc.edu.ph
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Project References

furlink system is design is guided by a combination of regional market analyses, localized consumer data, digital commerce standards, and comparative platform case studies. These references serve as the baseline requirements for resolving inefficiencies and visibility limitations identified within the local pet care ecosystem. The following references inform the platform’s architectural, data management, and user experience design decisions:

Philippine Pet Care Market Report (2025): Documents the expansion of the domestic pet care sector to a \$1.2 billion USD valuation, detailing the 22% annual growth rate in grooming services that justifies the technical scalability of the furlink database repository.

Vero ASEAN Consumer Behavior Survey (2024): Established that Filipino pet owners allocate an average of 15% of their household budgets to pet-related expenses, serving as the basis for the user interface metrics and target audience definitions.

Metro Manila Pet Service Provider Logistics Survey: Outlines the operational benchmarks of local grooming businesses, detailing how a 70% reliance on informal messaging channels

results in a 33% double-booking rate, which directly defines the core system requirements for furlink's automated calendar synchronization modules.

Platform UX Case Studies (Grab, Agoda, and Foodpanda): Provides the architectural blueprints for modular platform design, geo-location searching, and seamless multi-channel navigation layouts.

Conversational Commerce Framework (Salesforce, Infobip, and Bloomreach Standards): Establishes the technical logic, intent-parsing parameters, and dialogue flow methodologies required to construct the real-time interaction capabilities of the AI Booking Assistant.

Comparative B2B/B2C Platform Reviews (PetBacker, Rover, and DaySmart): Identifies functional gaps in existing market options—such as a lack of upfront pricing transparency and localized availability—which directly guide furlink's custom data flows for registration, instant quotation generation, and salon listing modules.

System Architecture

The architecture necessary to achieve the system design for furlink consists of following components:

Hardware

The furlink ecosystem is engineered as a fully cloud-based environment and rejects any dependency on physical or on-premises server hardware. Computing allocations, routing infrastructure, and file storage architectures are managed exclusively through specialized third-party cloud platforms.

- Vercel: Hosts and deploys the frontend web application with automated builds and continuous deployment connected to the GitHub Repository. It also serves as the authoritative domain registry and DNS management hub, anchoring all client traffic to the official system domain.

- Supabase: Manages the relational PostgreSQL database instances, handles core computing triggers, provisions secure cloud user authentication servers, and isolates encrypted binary file storage buckets.
- User Devices: The architecture restricts hardware dependencies entirely to the end-user side. The web platform is universally accessible via any modern consumer device running a standard, browser-capable application layer, completely removing the need for proprietary local hardware.

Software

Frontend

- React.js: Functions as the core foundational framework for client-side user interface construction, state management, and real-time interface updates.
- Vite: Utilized as the high-speed bundler and development server, executing highly optimized production code compilation and fast refresh modules to maximize development lifecycle speeds.
- Chart.js Library: Integrated directly into the provider interface to drive the interactive Business Dashboard, executing high-performance rendering of client-side canvas data visualizations to map sales trajectories, booking velocities, and provider revenue matrices.
- Vercel AI SDK: Implemented to establish streaming communication pipelines between the React presentation view and downstream artificial intelligence nodes, optimizing multi-turn chat performance, orchestrating local user prompt states, and receiving structured payload responses.

Backend

- Supabase Auth with JSON Web Token (JWT): Manages user session security by issuing cryptographically signed JWT keys upon successful login, verifying identities across subsequent transaction requests.

- Supabase Row Level Security (RLS): Enforces strict database-level access boundaries by automatically evaluating JWT claims against predefined SQL access policies, ensuring users only interact with their explicitly authorized data rows.
- Supabase Edge Functions: Handles serverless background logic including webhooks, AI orchestration, and automated triggers. The Gemini API key is stored securely as a Supabase secret and is never exposed to the client.
- Gmail SMTP: Delivers transactional email notifications triggered by Edge Functions.

Payments

- PayMongo: Payment gateway integrated via API for booking payment processing and webhook-based confirmation.

Artificial Intelligence

- Google Gemini 2.5 Flash: Powers both the conversational AI Booking Assistant and the analytical AI-Powered Data Analyst modules. This engine was selected for its high-speed transactional speed, cost-efficient token mapping, and extensive context window, which allows it to naturally retain comprehensive pet profiles and conversation histories across multi-turn booking dialogs or massive provider sales datasets.
- Vercel AI SDK with @ai-sdk/google provider: Orchestrates communication between the React frontend and the Gemini model, enabling tool calling for database queries and structured data extraction for form autofill, as well as a dynamic chart and report generation for the AI-Powered Data Analyst.
- Image Generation API: Powers the backend execution for the AI Pet Haircut Generator module, isolating intensive graphic rendering away from client-side resources. The architecture implements a tiered dual-provider configuration strategy handled via Supabase Edge Functions. The system establishes Hugging Face Serverless Inference as the primary graphic processing node to

generate visual pet style previews from uploaded user photographs. If the Hugging Face API triggers a transient rate limit, timeout, or quota restriction, the backend framework executes an automated failover handshake to route the image payload to Pollinations.ai as the backup generation container, ensuring high application uptime while shielding private vendor tokens from exposure.

Version Control, Governance, and Repository Frameworks

- **GitHub Repository and CI/CD Pipelines:** Functions as the absolute single source of truth for the furlink codebase. It enforces strict version control boundaries, protects development branches, and forms the core of the continuous integration and continuous deployment (CI/CD) infrastructure by triggering automated, production-ready builds on Vercel the moment an approved code modification is safely merged into the main development line.
- **GitHub Codespaces:** Serves as the team's unified, cloud-hosted virtual development infrastructure, providing containerized developer environments to ensure absolute runtime consistency across parallel programming sprints.

Software Modules

The following modules operate through the shared Supabase backend:

- Authentication and Access Control Module
- User Profile Management Module
- Pet Profile Module
- Service Provider Onboarding Module
- Platform Admin Oversight and Moderation Module
- Transaction and Payment Module
- Provider Operations Module
- Cancellation and Rescheduling Module
- Business Intelligence Dashboard Module

- AI Booking Assistant Module
- AI Pet Haircut Generator Module
- AI-Powered Data Analyst Module
- Discovery and Filtering Module

Database Design

The database layer of furlink is provisioned and hosted natively on Supabase, leveraging a cloud-based, high-performance PostgreSQL relational database engine. Functioning as the absolute single source of truth and centralized data repository for the entire platform, this layer guarantees that data inputs processed through any interface entry point are immediately consistent, normalized, and accessible across parallel software modules. To optimize data distribution and security, all database tables implement universally unique identifiers (UUIDs) as primary keys, eliminating sequential enumeration vulnerabilities and ensuring collision-free record generation. Furthermore, the entire schema architecture is strictly governed by PostgreSQL Row-Level Security (RLS) policies that enforce data access boundaries directly at the engine layer. These cryptographically checked RLS parameters validate user JSON Web Tokens (JWT) upon every transaction, ensuring that authenticated Pet Owners, Service Providers, and Application Administrators can execute data manipulation commands strictly within their pre-authorized boundaries. Because this data governance applies independently of the presentation view, any unauthorized structural operations or cross-tenant queries are rejected at the database level even if a malicious request successfully bypasses front-end application logic.

The database is organized into seven logical groups corresponding to the platform's core functional areas:

- Identity and Access
- Pet Management
- Service Provider
- Services and Pricing

- Booking Flow
- Post-Service
- Notifications

Each group contains tables that are directly linked through foreign key relationships to maintain referential integrity across the system. Several automated behaviors are enforced at the database level through PostgreSQL triggers and Edge Functions, including real-time email and in-app notifications, automatic booking decline after 24 hours of no service provider response, and automatic booking completion 3 days after the scheduled data to unlock the review window. The AI Booking Assistant also interacts with the database through secured Edge Functions, querying the tables for availability and information searches and writing when auto-filling and confirming bookings.

Hardware and Software Detailed Design

Hardware

furlink does not utilize or maintain any physical or on-premises server hardware, as the entire core infrastructure is deployed across third-party cloud service providers and managed serverless environments as documented in the System Architecture section.

Software

furlink's software architecture is organized into highly cohesive modules designed to isolate distinct boundaries of platform functionality. The system layout was engineered around the specific operational needs of pet owners, service providers, and application administrators, and was compartmentalized into decoupled modules to allow independent development, targeted testing, and parallel maintenance. All software modules communicate asynchronously through the shared Supabase backend engine and are supported by standalone architectural diagrams for detailed technical reference:

- **Authentication and Access Control Module:** Manages registration, login, and role assignment for three user types through Supabase Authentication and JWT, with permissions enforced through RBAC and RLS.
- **User Profile Management Module:** Allows users to update personal details, change passwords, view their role, and deactivate their account via soft-delete mechanism.
- **Pet Profile Module:** Enables pet owners to create and maintain permanent pet profiles that can be referenced during the booking process to reduce data entry.
- **Service Provider Onboarding Module:** An application workflow collecting business credentials, service offerings, pricing, staff, operating hours, and supporting documents for application admin review.
- **Platform Admin Oversight and Moderation Module:** Provides the application administrator with tools to approve or reject service provider applications and manage user warnings or suspensions.
- **Transaction and Payment Module:** Initiates a PayMongo checkout session upon booking, redirects the user to PayMongo's hosted page, and finalizes the booking only upon receiving a verified webhook confirmation. furlink does not handle raw card details at any point.
- **Provider Operations Module:** Gives service providers a real-time interface to accept or decline booking requests, with status changes reflected instantly on the pet owner's appointment page and booking history.
- **Cancellation and Rescheduling Module:** Handles cancellations, rescheduling requests, and post-service review submission, with review eligibility automatically unlocked three days after the scheduled booking date.
- **Business Intelligence Dashboard Module:** Presents service providers with visual summaries of business performance, sales performance, and customer insights.
- **AI Booking Assistant Module:** Implements a natural language interface powered by the Google Gemini 2.5 Flash API and orchestrated via the Vercel AI SDK to automate and simplify client scheduling workflows through three core capabilities:

- Knowledge Retrieval allows the conversational assistant to scan active marketplace data—including provider service listings, real-time availability matrices, and general platform FAQs—by triggering secured Supabase Edge Functions that execute read-only database queries and summarize the raw outputs into plain text for the user.
- Automated Data Entry interprets unstructured natural language chat streams to extract structured parameters (such as pet names, grooming selections, and date), automatically populating the hidden Pet Registration and Booking Forms for immediate user verification and manual confirmation.
- Payment Orchestration triggers a server-side Edge Function calling the external PayMongo API to generate a unique checkout link once the client confirms their data parameters. The assistant injects this secure URL directly into the chat stream for user completion, allowing the underlying database row to be safely updated only after a PayMongo webhook confirms successful clearing.
- **AI Pet Haircut Generator Module:** Available to pet owners during the booking pipeline if their selected service includes a styling haircut option. The component detects the pet's verified type and breed to return an explicit menu of standard haircut options. The user can select a style and optionally upload a photograph of their pet to receive an AI-generated preview illustrating how the styling configuration will look on their specific pet. This asset serves as an immutable visual reference linked directly to the booking record to optimize provider grooming accuracy. The generation logic is handled server-side via a Supabase Edge Function to protect the private access keys. The uploaded user photograph and the resulting generated preview are stored securely within the database rows as `ai_reference_url` and `ai_generated_url` respectively. To ensure maximum uptime, a dual-provider redundancy layer is engineered into the Edge Functions: the system primarily executes the request via a high-performance model hosted on a Hugging Face Inference Endpoint; if the primary endpoint returns an error, timeout, or quota restriction, an automated failover

instantly reroutes the payload to Pollinations.ai as the backup generation container, eliminating user friction during peak periods or third-party outages.

- **AI-Powered Data Analyst Module:** Integrated exclusively within the provider-side Business Intelligence Dashboard. The module introduces an interactive conversational interface powered by the Google Gemini 2.5 Flash API that allows service providers to execute natural language queries against their local operational metrics and receive analytical insights, custom data arrays, and structured performance trends in return. The module calls secured Supabase Edge Functions that are strictly sandboxed via custom SQL views, ensuring that each provider can only access and analyze records belonging to their own authenticated account, completely blocking cross-tenant data visibility. This module extends the existing Business Intelligence Dashboard Module by adding a dynamic, conversational layer on top of the static visual summaries already available.
- **Discovery and Filtering Module:** Allows pet owners to search and filter service providers by location, star rating, and price range.

System Security and Integrity Controls

furlink incorporates multiple layers of security and integrity controls through a multi-tiered approach covering authentication, authorization, financial tokenization, payment security, storage, data integrity, and operational continuity.

- **Authentication and Authorization:** Core access control is enforced through Supabase Auth utilizing cryptographically signed JSON Web Tokens (JWT). Upon successful identification, tokens are securely issued to the client browser, scoped to the individual's specific system role, and set to automatically expire after a defined session period. Role-Based Access Control (RBAC) rules restrict Pet Owners, Service Providers, and Administrators to their explicitly authorized platform modules. Simultaneously, Supabase Row-Level Security (RLS) policies evaluate these JWT parameters directly at the database engine layer to block unauthorized operations,

ensuring that suspended or deactivated user accounts are rejected regardless of frontend presentation states or token generation anomalies.

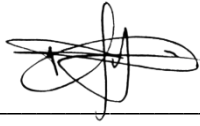
- **Input Validation:** Validation rules are enforced across all forms to prevent malformed or malicious data from entering the system. These rules cover required fields, data types, character limits, and format constraints, and are documented in the project's test cases.
- **Payment Security:** To guarantee absolute transactional compliance with global payment cards industry standards, furlink completely decouples its infrastructure from financial data handling by utilizing PayMongo's secure, hosted checkout ecosystem. Raw credit card numbers, banking credentials, and digital wallet keys are never collected, processed, or cached within the platform's database. Financial finalization remains strictly frozen until an automated, cryptographically signed webhook confirmation signal is transmitted from PayMongo's PCI-compliant servers to the Supabase Edge Functions. Under this framework, the conversational AI Booking Assistant is programmatically restricted from accessing or storing billing details, functioning solely as an encoder to render external transaction links.
- **AI Security:** Artificial intelligence orchestration keys are securely isolated away from public application code. The Google Gemini enterprise token, the Hugging Face Serverless Inference key, and all alternative Pollinations.ai access vectors are stored exclusively as encrypted Supabase environment secrets, executing entirely within server-side Edge Functions. To align with the Data Privacy Act of 2012, system prompt boundaries strictly blind the AI Booking Assistant from reading user logs outside of the active session context. Concurrently, the AI-Powered Data Analyst is securely sandboxed via isolated SQL views, ensuring its text-generation routines can only query aggregated historical telemetry belonging explicitly to the authenticated provider account, neutralizing lateral cross-tenant data leaks.
- **Storage and Media Security:** Pet media files including vaccine records, illness documents, pet reference photos uploaded for the AI Pet Haircut Generator, and AI-generated haircut preview images are stored in Supabase Storage and are planned

for migration to private bucket access, where files will only be accessible through time-limited signed URLs tied to authenticated user sessions.

- **Account Integrity:** User deactivation requests utilize a structured soft-delete mechanism that immediately flags the account status row as inactive and revokes database read-write tokens while preserving historical logs for transaction tracing and compliance audits. Complete account purging and irreversible database row deletions are executed strictly upon the submission of a formal data erasure request, maintaining complete alignment with the mandates of the Philippine Data Privacy Act of 2012 (R.A. 10173).
- **Audit Trail:** Login events are recorded in the `user_sessions` table, and database triggers across key tables log all significant data modifications, providing a traceable record of system activity for administrator review.
- **Operational Continuity:** Platform availability is secured through an infrastructure layout that balances vendor redundancy with local failover protocols. The core presentation layer and database schemas inherit the high-availability clustering, automated daily snapshot distributions, and routine background backup configurations built directly into Vercel and Supabase cloud topologies. For distributed AI features, continuity is split by operational impact: the single-model dependency on Google Gemini for conversational modules represents an accepted project risk backed by its 99.9% uptime, utilizing a frontend graceful degradation protocol to route users to standard database menus during rare outages. For the intensive graphic rendering pipeline, continuity is maintained through an active multi-model failover mechanism inside the Edge Functions that instantly reroutes traffic from a timed-out Hugging Face inference gateway to a backup Pollinations.ai container, minimizing user friction.

Sponsor Acceptance

Approved by the Project Sponsor:



Manuel R. Doria Jr.
Project Sponsor Representative

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